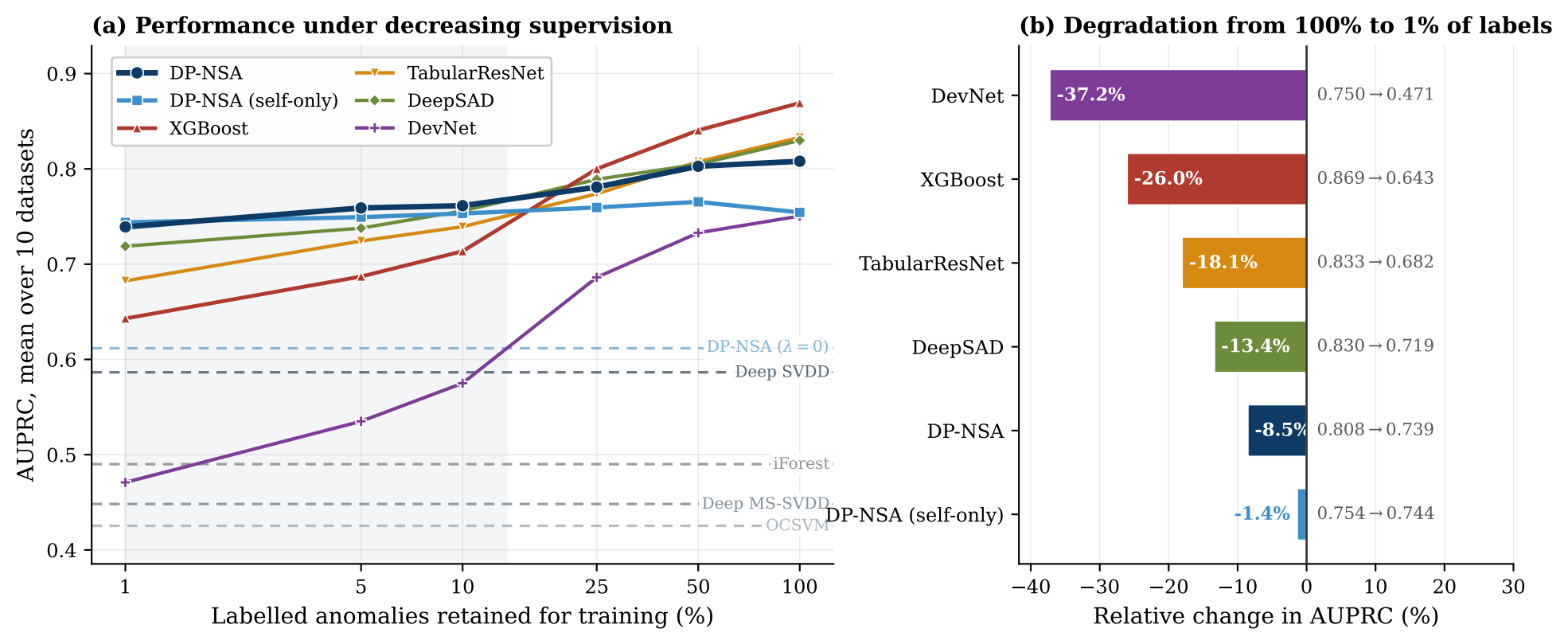
(a) Performance under decreasing supervision
DP-NSA
DP-NSA (self-only)
XGBoost
TabularResNet
DeepSAD
DevNet
AUPRC, mean over 10 datasets
0.9
0.8
0.7
0.6
0.5
0.4
DP-NSA (λ = 0)
Deep SVDD
iForest
Deep MS-SVDD
OCSVM
1
5
10
25
50
100
Labelled anomalies retained for training (%)
(b) Degradation from 100% to 1% of labels
DevNet
XGBoost
TabularResNet
DeepSAD
DP-NSA
DP-NSA (self-only)
-37.2%
-26.0%
-18.1%
-13.4%
-8.5%
-1.4%
0.750 → 0.471
0.869 → 0.643
0.833 → 0.682
0.830 → 0.719
0.808 → 0.739
0.754 → 0.744
−40
−30
−20
−10
0
10
20
30
Relative change in AUPRC (%)